Lecture 12

Transverse reduction

The Gauss constraint restricts Maxwell's momentum. Gauge transformations identify different spatial potentials. We impose the restriction and make the identification explicitly, using Fourier transformation away from zero frequency. The remaining variables have two independent directions at each nonzero wave vector.

Objectives. *We construct the transverse projector, identify the null directions of the constrained canonical form and derive the reduced bracket. Constant fields on a torus show why zero frequency requires a separate analysis.*

By the end of this lecture you will be able to:

- Find the unique transverse representative of an annulus-supported gauge orbit.
- Prove that the constrained form becomes nondegenerate after the gauge quotient.
- Compute the transverse smeared bracket and retain torus harmonic modes.

12.1 A class separated from zero frequency

Fix $0 < k_- < k_+ < \infty$ and the open Fourier annulus $\mathcal{U} = \{k \in \mathbb{R}^3 : k_- < |k| < k_+\}$. We use the unitary Fourier transform of Lecture 4.

Definition 12.1 (Annulus data class). The real *annulus data class* is

$$\mathcal{X} = \{f \in \mathcal{S}(\mathbb{R}^3, \mathbb{R}) : \widehat{f} \in C_c^\infty(\mathcal{U}, \mathbb{C})\}.$$

Its vector version is $\mathcal{X}^3$. All data, smearing functions and gauge parameters in the main reduction belong to these classes.

Reality means $\widehat{f}(-k) = \overline{\widehat{f}(k)}$. Inverse transforms of smooth compact frequency profiles are Schwartz functions, by the Fourier automorphism recalled in Lecture 7. Smooth multipliers on a neighbourhood of $\mathcal{U}$ preserve frequency support. The multipliers used below also preserve Fourier reality, so they preserve this real class. No division by $|k|$ below reaches $k = 0$.

We work in vacuum, $\rho = 0$. Lecture 11 gives $\mathbf{\Pi} = -\mathbf{E}$ and the Gauss constraint $\operatorname{div} \mathbf{\Pi} = 0$. The primary constraint sets $\pi_\phi = 0$, while its generator $P[\zeta]$ in (11.10) shifts the scalar potential arbitrarily. Removing this pair leaves the spatial data space and constraint surface given by

$$\mathcal{Z}_\mathcal{U} = \mathcal{X}^3 \times \mathcal{X}^3, \qquad C = \{(\mathbf{A}, \mathbf{\Pi}) \in \mathcal{Z}_\mathcal{U} : \operatorname{div} \mathbf{\Pi} = 0\}.$$

The constraint is equivalently $k \cdot \widehat{\mathbf{\Pi}}(k) = 0$. For $\chi \in \mathcal{X}$, its gauge action is $(\mathbf{A}, \mathbf{\Pi}) \mapsto (\mathbf{A} + \nabla\chi, \mathbf{\Pi})$. Its generator has the positive-gradient convention of (11.15): $G[\chi] = \int \mathbf{\Pi} \cdot \nabla\chi \, \mathrm{d}^3x$.

Definition 12.2 (Gauge orbit and reduced phase space)**.** The *gauge orbit* of $(\mathbf{A}, \mathbf{\Pi}) \in C$ is

$$[(\mathbf{A}, \mathbf{\Pi})] = \{(\mathbf{A} + \nabla\chi, \mathbf{\Pi}) : \chi \in \mathcal{X}\}.$$

The *reduced phase space* $C/\mathcal{G}$ is the set of these orbits, where $\mathcal{G}$ denotes the additive group of gauge parameters.

The canonical form from Lecture 8, summed over the three spatial components, is the bilinear pairing

$$\Omega((a, p), (b, q)) = \int_{\mathbb{R}^3} (a \cdot q - b \cdot p)\, \mathrm{d}^3x. \tag{12.1}$$

Its restriction to C can have a kernel. A vector belongs to this kernel when its pairing with every allowed tangent vector is zero. Here tangent vectors to C are pairs (a, p) with $\operatorname{div} p = 0$.

12.2 Projection and the quotient

Definition 12.3 (Transverse projector)**.** For $k \neq 0$, the *transverse projector* is the matrix

$$P_{\mathrm{T}}(k) = I - \frac{kk^{\mathsf{T}}}{|k|^2}, \qquad P_{\mathrm{T}}(k)_{ij} = \delta_{ij} - \frac{k_i k_j}{|k|^2}. \tag{12.2}$$

For $v \in \mathcal{X}^3$, define its transverse part by $\widehat{v_T}(k) = P_{\mathrm{T}}(k)\widehat{v}(k)$. We also write $P_{\mathrm{T}}v = v_T$ for this spatial operator. A vector field is *transverse* when $\operatorname{div} v = 0$.

The matrix projects a vector onto the plane perpendicular to k. Figure 12.1 separates this transverse part from the longitudinal part parallel to k.

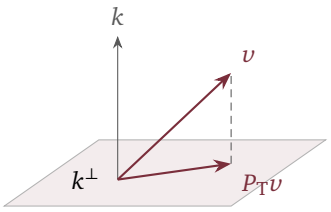


Figure 12.1: The transverse plane $k^{\perp}$. A vector splits as $v = P_{\mathrm{T}}v + (I - P_{\mathrm{T}})v$, with the second component parallel to k.

For real smearing functions $f, g \in \mathcal{X}^3$, the reduced linear observables are

$$A_T(f) = \int f \cdot \mathbf{A}_T\, \mathrm{d}^3x, \qquad \Pi_T(g) = \int g \cdot \mathbf{\Pi}_T\, \mathrm{d}^3x.$$

We consider finite polynomials in these observables. Their gradients remain in the annulus class. This specifies the observable algebra used for the reduced bracket.

Theorem 12.4 (Transverse reduction). *In the stated annulus class, every orbit in $\mathcal{C}$ has exactly one representative with* $\operatorname{div}\mathbf{A}_T = 0$. *The quotient is identified with*

$$\mathcal{Z}_T = \{(\mathbf{A}_T, \mathbf{\Pi}_T) \in \mathcal{X}^3 \times \mathcal{X}^3 : \operatorname{div}\mathbf{A}_T = \operatorname{div}\mathbf{\Pi}_T = 0\}.$$

The kernel of the constrained canonical form consists exactly of the gauge directions $(\nabla\chi, 0)$. *The form therefore descends to the nondegenerate canonical form* (12.1) *on* $\mathcal{Z}_T$. *The reduced bracket on smeared generators is*

$$\{A_T(f), \Pi_T(g)\}_{\mathrm{red}} = \int f \cdot P_{\mathrm{T}} g \, \mathrm{d}^3x, \qquad \{A_T(f), A_T(g)\}_{\mathrm{red}} = \{\Pi_T(f), \Pi_T(g)\}_{\mathrm{red}} = 0. \tag{12.3}$$

The bracket on their polynomial algebra follows by bilinearity and the product rule. At each $k \neq 0$, *the transverse position and momentum each have two independent components.*

Proof. First, put $K = kk^{\mathsf{T}}/|k|^2$. Since $k^{\mathsf{T}}k = |k|^2$, $K^2 = K$. Thus $P_{\mathrm{T}}^2 = (I - K)^2 = I - K = P_{\mathrm{T}}$. Also $P_{\mathrm{T}}^{\mathsf{T}} = P_{\mathrm{T}}$, $P_{\mathrm{T}}k = 0$ and $\operatorname{tr} P_{\mathrm{T}} = 3 - 1 = 2$. These identities show that P_{T} is the orthogonal projection onto $k^{\perp}$. It is real and even in k, so it preserves Fourier reality.

The transverse part of a potential is gauge invariant because $P_{\mathrm{T}}(\widehat{\mathbf{A}} + ik\widehat{\chi}) = P_{\mathrm{T}}\widehat{\mathbf{A}}$. To reach that representative, choose

$$\widehat{\chi}(k) = \frac{i\,k \cdot \widehat{\mathbf{A}}(k)}{|k|^2}.$$

This is a smooth compact annulus profile with the required reality condition. It gives $\widehat{\mathbf{A}} + ik\widehat{\chi} = P_{\mathrm{T}}\widehat{\mathbf{A}}$. If two transverse representatives differ by $ik\widehat{\chi}$, taking the dot product with k gives $i|k|^2\widehat{\chi} = 0$. Thus they coincide. The constraint already gives $\mathbf{\Pi} = \mathbf{\Pi}_T$.

Next, a gauge direction pairs to zero with every $(b, q) \in \mathcal{C}$:

$$\Omega((\nabla\chi, 0), (b, q)) = \int \nabla\chi \cdot q \, \mathrm{d}^3x = -\int \chi \, \operatorname{div} q \, \mathrm{d}^3x = 0.$$

Schwartz decay removes the face terms, as in Lecture 7. Conversely, suppose (a, p) is in the kernel. Testing against $(p, 0)$ gives $-\int |p|^2 = 0$, so $p = 0$. Testing against $(0, P_{\mathrm{T}}a)$ gives $\int a \cdot P_{\mathrm{T}}a = \int |P_{\mathrm{T}}a|^2 = 0$, by symmetry and idempotence of the multiplier and Plancherel. Hence $P_{\mathrm{T}}a = 0$, and the preceding gauge construction expresses a as a gradient. Removing these null directions gives a well-defined form on orbits. For a transverse pair (a, p), its pairing with $(-p, a)$ is $\|a\|_2^2 + \|p\|_2^2$. Thus the quotient form is nondegenerate. It is closed because its coefficients are constant.

Finally, symmetry of P_{T} in the real L^2 pairing gives $A_T(f) = \int (P_{\mathrm{T}}f) \cdot \mathbf{A}$ and $\Pi_T(g) = \int (P_{\mathrm{T}}g) \cdot \mathbf{\Pi}$. Their canonical gradients are $(P_{\mathrm{T}}f, 0)$ and $(0, P_{\mathrm{T}}g)$. The bracket of Lecture 7 therefore gives

$$\{A_T(f), \Pi_T(g)\} = \int P_{\mathrm{T}}f \cdot P_{\mathrm{T}}g \, \mathrm{d}^3x = \int f \cdot P_{\mathrm{T}}g \, \mathrm{d}^3x.$$

These observables are constant on each gauge orbit, so this is their reduced bracket. Their Hamiltonian vectors are transverse and agree with those obtained directly from the quotient form. The same construction and the product rule apply to their finite polynomials. ∎

12.3 Two polarisations and the Fourier bracket

A *polarisation vector* at $k \neq 0$ is a unit vector in $k^{\perp}$. Choose an orthonormal pair $\varepsilon_1(k)$ and $\varepsilon_2(k)$ at that fixed k. The two components of $\widehat{\mathbf{A}}_T$ along this pair, and the two corresponding momentum components, describe the transverse data there. This pointwise choice does not require a smooth basis on the whole annulus.

The Fourier transform of a real field couples k to $-k$ through its reality condition. On smooth frequency tests supported in $\mathcal{U}$, the bracket identity can be written in the formal kernel notation

$$\{\widehat{A}_{T,i}(k), \widehat{\Pi}_{T,j}(k')\}_{\mathrm{red}} = P_{\mathrm{T}}(k)_{ij}\, \delta(k + k'). \tag{12.4}$$

This notation means that testing with profiles $\alpha_i(k)$ and $\beta_j(k')$ gives $\int \alpha_i(k) P_{\mathrm{T}}(k)_{ij} \beta_j(-k)\, \mathrm{d}^3k$. Complex tests use the complex bilinear extension of the real bracket. This specifies only the tested identity on the chosen annulus class. The sum $k + k'$ comes from the two forward Fourier phases:

$$(2\pi)^{-3} \int_{\mathbb{R}^3} e^{-i(k+k')\cdot x}\, \mathrm{d}^3x = \delta(k + k').$$

As in Lecture 7, the kernel notation does not make mode point values admissible observables.

Example 12.5 (The bracket along a coordinate axis)**.** Take $k = (0, 0, k_3) \in \mathcal{U}$, with $k_3 \neq 0$. The projector and transverse Fourier data are

$$P_{\mathrm{T}}(k) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \widehat{\mathbf{A}}_T = (\widehat{A}_1, \widehat{A}_2, 0), \qquad \widehat{\mathbf{\Pi}}_T = (\widehat{\Pi}_1, \widehat{\Pi}_2, 0).$$

Choose $\varepsilon_1 = e_1$ and $\varepsilon_2 = e_2$. In the tested notation of (12.4), the two-component bracket matrix at this wave vector is

$$\big(\{\widehat{A}_\alpha(k), \widehat{\Pi}_\beta(k')\}_{\mathrm{red}}\big)_{\alpha,\beta=1}^{2} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \delta(k + k').$$

The Gauss constraint removes $\widehat{\Pi}_3$, and gauge equivalence removes $\widehat{A}_3$. It leaves two canonical pairs, with Fourier reality relating the coefficients at the opposite wave vector.

12.4 The zero mode on the torus

The annulus assumption excludes harmonic data. On $\mathbb{T}^3$, a constant vector field is *harmonic*: every component has zero Laplacian. Take constant data $\mathbf{A} = a$ and $\mathbf{\Pi} = p$, with $a, p \in \mathbb{R}^3$. The Gauss constraint holds automatically. Conversely, $\Delta v = 0$ on the torus gives $|n|^2 \widehat{v}_n = 0$, so all harmonic Fourier components have $n = 0$ and are constant. A periodic gauge parameter obeys $\int_{\mathbb{T}^3} \nabla\chi\, \mathrm{d}^3x = 0$, by opposite-face cancellation. Consequently it cannot change the mean a. These constant modes survive the quotient, with canonical form

$$\Omega_{\mathrm{harm}} = (2\pi)^3 \sum_{i=1}^{3} \delta a_i \wedge \delta p_i.$$

There are three harmonic pairs. Applying the nonzero-k count to $k = 0$ would incorrectly remove one of them. Here gauge parameters are single-valued periodic real functions, as in the gauge action above. Lectures 20 and 21 must retain harmonic and global data when the spatial topology changes.

12.5 The bounded projector and the energy-space scope

For any $v \in L^2(\mathbb{R}^3;\mathbb{R}^3)$, define $P_{\mathrm{T}}v$ directly by the multiplier (12.2) for $k \neq 0$. Its value at the single point $k = 0$ has no effect on an L^2 function. Orthogonality gives

$$\|P_{\mathrm{T}}v\|_2^2 = \int |P_{\mathrm{T}}(k)\widehat{v}(k)|^2 \,\mathrm{d}^3k \leq \int |\widehat{v}(k)|^2 \,\mathrm{d}^3k = \|v\|_2^2.$$

Thus this is a bounded orthogonal projector on all of L^2, not merely on one fixed annulus. The same calculation with Fourier weight $1 + |k|^2$ gives its bounded action on H^1.

The corresponding *inhomogeneous energy-space reduction* also follows from this projector. Take $\mathbf{A} \in H^1$ and $\mathbf{\Pi} \in L^2$ with $\operatorname{div}\mathbf{\Pi} = 0$ distributionally, and quotient by distributional gradients $\nabla\chi \in H^1$. The reduced space is

$$H^1_{\mathrm{trans}}(\mathbb{R}^3;\mathbb{R}^3) \times L^2_{\mathrm{trans}}(\mathbb{R}^3;\mathbb{R}^3),$$

where the subscript means distributional divergence zero. Indeed, set $\widehat{\chi} = -i(k \cdot \widehat{\mathbf{A}})/|k|^2$ for $k \neq 0$. This defines a tempered distribution: near zero, Cauchy–Schwarz uses $\int_{|k|<1} |k|^{-2}\mathrm{d}^3k < \infty$ to prove local integrability; at infinity its pairing with every Schwartz test converges by Cauchy–Schwarz. It preserves Fourier reality and gives $\nabla\chi = (I - P_{\mathrm{T}})\mathbf{A} \in H^1$. Conversely, an H^1 gradient is longitudinal: its Fourier curl is zero, so $k \times \widehat{\nabla\chi} = 0$ almost everywhere and its transverse projection vanishes. Thus each orbit has exactly one transverse representative. The canonical form descends because longitudinal gradients are L^2-orthogonal to transverse momenta, by Plancherel.

Its form (12.1) is weakly nondegenerate. If (a, p) pairs to zero with every transverse pair, testing with $(0, a)$ first gives $a = 0$. For the momentum test take $(h, 0)$, where $\widehat{h} = (1 + |k|^2)^{-1}\widehat{p}$. Then h is transverse and $\|h\|_{H^1}^2 = \int (1 + |k|^2)^{-1}|\widehat{p}|^2\mathrm{d}^3k \leq \|p\|_2^2$, and the pairing is $-\int (1 + |k|^2)^{-1}|\widehat{p}|^2\mathrm{d}^3k$. Its vanishing gives $p = 0$. The form is closed because it is constant. Here *weak* means nondegenerate without requiring every continuous functional to have a representing vector. For the resulting transverse Sobolev phase model, see Emberton, *An introduction to free quantum field theory through Klein–Gordon theory*, §7, p. 14, in the University of Chicago notes. Their momentum is $\mathbf{E}$, while ours is $\mathbf{\Pi} = -\mathbf{E}$. The fixed-annulus theorem above remains independent of this broader argument. Sheet 12 develops the completion and gauge-parameter details further. Requiring only finite physical energy controls $\operatorname{curl}\mathbf{A}$ and $\mathbf{\Pi}$, but need not put $\mathbf{A}$ in L^2. It would not by itself justify the canonical integral used here.

Summary. For vacuum annulus-supported Schwartz data, the Gauss constraint sets the longitudinal momentum to zero and gauge equivalence removes the longitudinal potential. Their quotient has the nondegenerate canonical form on $(\mathbf{A}_T, \mathbf{\Pi}_T)$ and the smeared bracket $\int f \cdot P_{\mathrm{T}}g$. At nonzero k, the projector has rank two. The tested Fourier bracket contains $\delta(k+k')$. On the torus, three constant canonical pairs survive periodic gauge transformations. The projector itself is bounded on all of L^2.

Self-check. At $k = (0, 0, 2)$, project $v = (3, -1, 5)$. Which canonical components does the Gauss constraint remove, and which does gauge equivalence remove?

Answer. The projection is $(3, -1, 0)$. The constraint removes the third momentum component. Gauge equivalence removes the third potential component.

Self-check. For annulus data, find the gauge parameter that takes $\mathbf{A}$ to $\mathbf{A}_T$. Why does it preserve the real-field condition?

Answer. Its transform is $\widehat{\chi} = i(k \cdot \widehat{\mathbf{A}})/|k|^2$. At $-k$, its value is $-i(k \cdot \overline{\widehat{\mathbf{A}}})/|k|^2$, the complex conjugate of its value at k.

Self-check. For a real nonzero Fourier mode, should the bilinear canonical bracket pair $\widehat{A}(k)$ with $\widehat{\Pi}(k)$ or with $\widehat{\Pi}(-k)$?

Answer. It pairs opposite wave vectors. The tested kernel contains $\delta(k + k')$, as follows from the two forward Fourier phases. A conjugated Hilbert inner product has a different frequency bookkeeping rule.

Self-check. On $\mathbb{T}^3$, can a periodic real gauge parameter remove a nonzero constant potential? What is the symplectic coefficient for each constant pair?

Answer. No. Its gradient has zero mean, so the constant potential is unchanged. Each pair has coefficient $\mathrm{vol}(\mathbb{T}^3) = (2\pi)^3$ in the canonical form.